

Airbnb Melbourne Data Engineering and Market Analysis Report

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1. Executive Summary

This project analyses the Melbourne, Victoria, Australia Airbnb market using public data from Inside Airbnb.

The objective was to build a reproducible data engineering and analytics workflow that transforms raw Airbnb files into cleaned, profiled, and analytics-ready outputs. The project includes dataset familiarisation, data profiling, cleaning, an enriched listing master table, an analytical DuckDB database, SQL analysis, exploratory data analysis, statistical testing, machine learning listing segmentation, and lightweight NLP review analysis.

A key dataset limitation was identified during profiling: the Melbourne Inside Airbnb files used in this project did not contain usable price values. The price field in the detailed listings file, the summary listings file, and the calendar file contained no non-null values. For this reason, the final analysis focuses on supply, availability, host concentration, review behaviour, neighbourhood patterns, quality signals, listing segmentation, and review themes rather than price or revenue.

Key findings include:

- Entire homes/apartments dominate Melbourne Airbnb supply, with 17,703 listings.
- Private rooms represent the second-largest room type, with 6,563 listings.
- The Melbourne neighbourhood has the highest listing count, with 7,832 listings.
- Darebin has the highest estimated occupancy-rate proxy, at 0.7066.
- Flexistayz is the largest host by listing count, managing 292 listings.
- Superhost status is statistically associated with different review score distributions.
- Availability patterns differ across room types, neighbourhoods, and weekend/weekday calendar records.
- K-Means clustering identified four behavioural listing segments, including low-availability active supply, high-availability idle supply, established high-review listings, and low-rating low-activity listings.
- Review NLP analysis found that guest review language is strongly positive overall, with common themes around location, cleanliness, host quality, comfort, and ease of stay.

The project intentionally focuses on one city to prioritise depth, code quality, reproducibility, and clear business interpretation.

2. Objectives and Scope

2.1 Objectives

The main objectives of this project were to:

- Understand the Inside Airbnb Melbourne dataset.
- Profile raw data quality.
- Clean and standardise key fields.
- Create an enriched listing-level analytical dataset.
- Build a simple analytical database model.
- Run SQL-based business analysis.
- Generate EDA outputs and visualisations.
- Apply statistical testing to support analytical findings.
- Implement an explainable ML segmentation experiment using available listing features.
- Apply lightweight NLP analysis to guest review text.
- Document assumptions, limitations, and technical decisions clearly.

2.2 Scope

This project focuses on one city:

Melbourne, Victoria, Australia

A single-city scope was selected because the assignment values depth and quality over attempting every possible task superficially.

2.3 Out of Scope

The following items were intentionally not completed:

- Multi-city comparison
- Machine learning price prediction
- Advanced transformer-based NLP modelling
- LLM-based review analysis or RAG system
- Production-grade model deployment
- Cloud deployment
- Dashboard deployment
- Docker containerisation

These items were deprioritised to focus on core data engineering, reproducibility, EDA, statistical analysis, ML segmentation, lightweight NLP review analysis, and clear documentation.

3. Dataset Overview

3.1 Dataset Source

The dataset was sourced from Inside Airbnb.

Files used:

- listings.csv.gz
- listings.csv
- calendar.csv.gz
- reviews.csv.gz
- neighbourhoods.csv
- neighbourhoods.geojson

3.2 Main Entities

The main entities in the dataset are:

- Listings: Airbnb properties available in the Melbourne market.
- Hosts: People or organisations managing one or more listings.
- Calendar records: Daily availability records for each listing.
- Reviews: Guest review records linked to listings.
- Neighbourhoods: Geographic groupings used for location-based analysis.

3.3 Key Relationships

The main dataset relationships are:

```
listings.id = calendar.listing_id  
listings.id = reviews.listing_id  
listings.neighbourhood_cleansed = neighbourhoods.neighbourhood  
listings.host_id identifies hosts
```

3.4 Dataset Limitation: Missing Price Values

During profiling, the Melbourne dataset was found to have no usable price data in the relevant files:

- listings.csv.gz price values were fully missing.
- listings.csv price values were fully missing.
- calendar.csv.gz price and adjusted price values were fully missing.

Because of this limitation, price and revenue analysis were excluded from the final analytical findings. This decision avoids producing unsupported or misleading conclusions.

The final analysis therefore focuses on:

- Listing supply
- Annual availability
- Calendar availability
- Review behaviour
- Review scores
- Host concentration
- Neighbourhood-level supply patterns
- Superhost quality signals
- Listing segmentation
- Review sentiment and review themes

4. Methodology

The project followed a structured data engineering and analytics workflow:

1. Raw data validation
2. Data profiling
3. Cleaning and standardisation
4. Feature derivation
5. Dataset enrichment
6. Analytical database modelling
7. SQL analysis
8. Exploratory data analysis
9. Statistical testing
10. Machine learning listing segmentation
11. Review NLP sentiment and theme analysis

12. Documentation and reporting

The full pipeline can be executed with:

```
python src/run_pipeline.py
```

5. Engineering Approach

5.1 Pipeline Design

The project uses a repeatable Python pipeline controlled by:

```
python src/run_pipeline.py
```

The pipeline runs the following scripts:

1. check_data.py
2. profile_data.py
3. clean_data.py
4. build_master_table.py
5. build_database.py
6. run_sql_analysis.py
7. create_eda_outputs.py
8. create_statistical_analysis.py
9. create_ml_experiments.py
10. add_segment_labels.py
11. create_ai_nlp_experiments.py
12. create_report_insights.py

This design keeps each stage modular while still allowing the full workflow to be regenerated with one command.

5.2 Data Cleaning

Cleaning steps included:

- Parsing date fields.
- Converting percentage fields into decimal format.
- Standardising text fields.
- Validating latitude and longitude values.
- Creating host tenure fields.
- Creating review frequency fields.
- Creating calendar-based weekend and weekday indicators.
- Preserving missing price values rather than imputing unsupported prices.

5.3 Enriched Listing Master Table

The enriched listing_master table combines listing attributes with calendar and review summaries.

Derived fields include:

- available_days
- unavailable_days
- occupancy_rate_estimate
- review_frequency_per_month
- listing_count_neighbourhood
- average_rating_neighbourhood

The occupancy_rate_estimate field is treated as a proxy because unavailable calendar days may represent bookings, blocked dates, maintenance periods, or host-disabled dates.

5.4 Analytical Database

DuckDB was used as the analytical database.

Created tables include:

- listing_master
- clean_calendar
- clean_reviews
- dim_host
- dim_neighbourhood
- dim_room_type
- dim_date
- fact_listing
- fact_calendar_daily
- fact_reviews

This provides a simple analytical star-schema style model for querying listing, calendar, and review behaviour.

5.5 ML and AI/NLP Outputs

The project also generates additional outputs for the ML and AI/ML components:

```
reports/ml_outputs/  
reports/ai_outputs/  
reports/figures/listing_segments_pca.png  
reports/figures/segment_profile_availability.png  
reports/figures/review_sentiment_distribution.png  
reports/figures/top_review_terms.png
```

These outputs extend the project beyond descriptive analysis by adding behavioural segmentation and review-text analysis.

6. Data Quality Findings

A data profiling report was generated at:

```
reports/data_quality_report.csv
```

The raw dataset sizes were:

- Listings: 24,491 rows
- Calendar: 8,939,215 rows
- Reviews: 943,744 rows
- Neighbourhoods: 30 rows

The most important data quality finding was that price fields were not usable. The detailed listings file, summary listings file, and calendar file all had missing price values. This directly affected the analytical scope and required the project to focus on non-price market intelligence.

This is a material limitation, but documenting it clearly strengthens the reliability of the analysis. The project avoids unsupported price or revenue conclusions and instead analyses fields with sufficient data quality.

7. SQL Analysis Findings

SQL analysis outputs were generated in:

```
reports/sql_outputs/
```

7.1 Listing Supply by Room Type

The largest room type category was Entire Home/Apt, with 17,703 listings. This category had an average annual availability of 151.37 days and an average of 46.39 reviews per listing.

Private rooms were the second-largest category, with 6,563 listings, average annual availability of 139.74 days, and an average of 18.45 reviews per listing.

This indicates that the Melbourne Airbnb market is strongly shaped by entire-home/apartment supply rather than shared accommodation.

7.2 Neighbourhood Supply

The neighbourhood with the highest listing count was Melbourne, with 7,832 listings. Its average annual availability was 140.12 days and the average number of reviews was 49.14.

This suggests that neighbourhood-level supply concentration is important for market monitoring and operational planning.

7.3 Estimated Occupancy Proxy

The neighbourhood with the highest estimated occupancy-rate proxy was Darebin, based on 584 listings. Its average occupancy-rate estimate was 0.7066, while average annual availability was 106.91 days.

This should be treated as a proxy because calendar unavailability may represent bookings, blocked dates, or host-disabled dates.

7.4 Host Concentration

The largest host by listing count was Flexistayz, managing 292 listings. This host had average annual availability of 94.66 days and average reviews per listing of 5.80.

This suggests that host concentration is an important supply-side feature to monitor, especially when distinguishing casual hosts from professional or high-volume operators.

8. Exploratory Data Analysis Findings

EDA outputs were generated in:

```
reports/eda_outputs/  
reports/figures/
```

Generated figures include:

- listing_supply_by_room_type.png
- availability_by_room_type.png
- top_neighbourhoods_by_listing_count.png
- review_score_distribution.png
- reviews_by_room_type.png
- monthly_availability_rate.png
- top_hosts_by_listing_count.png
- listing_segments_pca.png
- segment_profile_availability.png
- review_sentiment_distribution.png
- top_review_terms.png

8.1 Listing Supply by Room Type

Figure: reports/figures/listing_supply_by_room_type.png

Business interpretation:

Entire homes/apartments dominate Melbourne Airbnb supply. This suggests the platform is not only serving shared accommodation use cases, but also a large whole-property short-term rental market. Market analysts should therefore evaluate Melbourne Airbnb activity as part of the broader accommodation and housing supply discussion.

8.2 Availability by Room Type

Figure: reports/figures/availability_by_room_type.png

Business interpretation:

Availability differs by room type, with shared rooms and hotel rooms showing higher average annual availability than entire homes and private rooms. Higher availability may indicate weaker demand, more flexible supply, or listings that are not actively booked throughout the year.

8.3 Top Neighbourhoods by Listing Count

Figure: reports/figures/top_neighbourhoods_by_listing_count.png

Business interpretation:

The Melbourne neighbourhood has the highest listing concentration. This indicates that Airbnb supply is spatially concentrated rather than evenly distributed across the city. Localised monitoring is therefore more useful than relying only on city-wide averages.

8.4 Review Score Distribution

Figure: reports/figures/review_score_distribution.png

Business interpretation:

Review score distribution helps assess customer satisfaction patterns. If scores are concentrated near the top end, small differences in rating may still matter commercially because many listings are competing within a narrow high-score band.

8.5 Reviews by Room Type

Figure: reports/figures/reviews_by_room_type.png

Business interpretation:

Entire homes/apartments have a higher average number of reviews than private rooms. This may indicate stronger guest activity, longer operating history, or greater visibility for entire-home listings.

8.6 Monthly Availability

Figure: reports/figures/monthly_availability_rate.png

Business interpretation:

Monthly availability patterns help identify potential seasonal or operational changes. Lower availability may suggest stronger demand or more host-blocked dates, while higher availability may indicate weaker demand or excess supply.

8.7 Host Concentration

Figure: reports/figures/top_hosts_by_listing_count.png

Business interpretation:

The host concentration chart identifies high-volume operators. This is useful for understanding whether supply is driven mainly by many casual hosts or by a smaller group of professional operators.

9. Statistical Findings

Statistical outputs were generated at:

```
reports/statistical_outputs/statistical_tests_summary.csv
```

Non-parametric tests were used because Airbnb marketplace variables such as availability, reviews, and review scores are unlikely to follow ideal normal distributions.

H1: Entire-home listings have different annual availability than private rooms

Test used: Mann-Whitney U test

- Sample size group 1: 17,703
- Sample size group 2: 6,563
- p-value: 0.000000
- effect size: 0.0659

Interpretation:

The result suggests that entire-home listings and private rooms have statistically different annual availability distributions. The effect size is small, so the practical difference should be interpreted carefully. However, it still indicates that accommodation type is relevant when analysing supply strategy.

H2: Superhost listings have different review scores than non-superhost listings

Test used: Mann-Whitney U test

- Sample size group 1: 6,781
- Sample size group 2: 12,735
- p-value: 0.000000
- effect size: 0.2590

Interpretation:

The result suggests that superhost status is associated with different review score distributions. The effect size is larger than the other tests in this project, indicating that superhost status may be a meaningful quality signal.

H3: Annual availability differs across neighbourhoods

Test used: Kruskal-Wallis test

- Total sample size: 24,491
- Neighbourhood groups: 30
- p-value: 0.000000
- effect size: 0.0565

Interpretation:

The result suggests that availability patterns differ across Melbourne neighbourhoods. The effect size is small, but the result supports neighbourhood-level monitoring rather than relying only on city-level averages.

H4: Weekend availability differs from weekday availability

Test used: Mann-Whitney U test

- Weekend records: 2,547,064
- Weekday records: 6,392,151
- p-value: 0.000000
- effect size: -0.0101

Interpretation:

The result suggests a statistically detectable difference between weekend and weekday availability. However, the effect size is very small, so the practical business impact may be limited. This is an example where statistical significance does not necessarily mean strong practical significance.

10. Data Science & ML Experiment: Listing Segmentation

To extend the analysis beyond descriptive reporting, I implemented a K-Means clustering experiment to segment Melbourne Airbnb listings into behavioural supply groups. The model used listing-level features including annual availability, estimated occupancy rate, review count, reviews per month, review score, host tenure, host listing count, room type, and superhost status.

The model produced four listing segments with a silhouette score of 0.3168. This indicates moderate separation between clusters, which is reasonable for real-world marketplace data where listing behaviour often overlaps rather than forming perfectly distinct groups.

The four listing segments were:

- Segment 0 - Low-availability active supply: 12,332 listings. These listings have low annual availability and a high occupancy proxy, suggesting stronger utilisation.
- Segment 1 - High-availability casual or idle supply: 8,483 listings. These listings have high annual availability and a lower occupancy proxy, suggesting casual, seasonal, or under-utilised supply.
- Segment 2 - Established high-review listings: 2,804 listings. These listings have very high review counts and strong review scores, suggesting established demand signals.
- Segment 3 - Low-rating low-activity listings: 872 listings. These listings have lower review scores, low review counts, and relatively high availability, suggesting potential quality or optimisation issues.

This segmentation creates a more practical market view than analysing listings only by room type or neighbourhood. For example, platform teams could prioritise Segment 3 for quality improvement support, while revenue or market teams could study Segment 2 to understand characteristics of established high-performing listings.

The experiment was intentionally framed as unsupervised segmentation rather than price prediction because the Melbourne dataset did not contain usable price values. This avoided creating unsupported predictive targets and kept the ML work aligned with validated data.

Key ML outputs were generated in:

```
reports/ml_outputs/  
reports/figures/listing_segments_pca.png  
reports/figures/segment_profile_availability.png
```

11. AI/ML Experiment: Review NLP Sentiment and Theme Analysis

To explore the unstructured review text, I implemented a lightweight NLP experiment using guest review comments. The analysis used text cleaning, lexicon-based sentiment scoring, and TF-IDF keyword extraction. This approach was selected because it is transparent, reproducible, and does not require external API keys or paid AI services.

The sentiment analysis classified review language into three categories:

- Positive: 809,852 reviews, representing 89.6% of analysed reviews.
- Neutral: 83,959 reviews, representing 9.3% of analysed reviews.
- Negative: 9,590 reviews, representing 1.1% of analysed reviews.

The results show that Melbourne Airbnb reviews are strongly skewed toward positive language. This aligns with the generally high review score distribution observed elsewhere in the project, but it also highlights a limitation: review sentiment alone may not be enough to identify listing quality issues because public review behaviour is highly positive overall.

The top review terms included great, stay, place, location, apartment, clean, host, comfortable, recommend, and great location. These terms suggest that guests most frequently describe the stay experience through location, cleanliness, host interaction, comfort, and ease of stay.

From a business perspective, this NLP analysis can support host success and operations teams by identifying the themes guests mention most often. It also provides a foundation for future work such as topic modelling, review-quality classification, or listing improvement recommendations based on guest language.

This AI/ML experiment was kept deliberately simple and explainable. A more advanced version could use transformer-based sentiment models or topic modelling, but the current approach is easier to audit and reproduce within the assessment scope.

Key AI/NLP outputs were generated in:

```
reports/ai_outputs/  
reports/figures/review_sentiment_distribution.png  
reports/figures/top_review_terms.png
```

12. Business Recommendations

Recommendation 1: Monitor Entire-Home Supply Closely

Entire homes/apartments dominate the Melbourne Airbnb dataset. Stakeholders should monitor this category separately because it represents the largest share of supply and may behave differently from private rooms or shared rooms.

Recommendation 2: Use Neighbourhood-Level Market Monitoring

The Melbourne neighbourhood has the highest listing concentration, while other areas such as Darebin show strong occupancy-rate proxy patterns. Market analysis should therefore be performed at neighbourhood level rather than relying only on city-wide averages.

Recommendation 3: Treat Occupancy Estimates Carefully

Occupancy estimates should be used as directional indicators only. Calendar unavailability may include actual bookings, host-blocked dates, maintenance, or inactive dates.

Recommendation 4: Track High-Volume Hosts

The largest host, Flexistayz, manages 292 listings. Host concentration analysis can help identify professional operators and distinguish them from casual hosts.

Recommendation 5: Use Superhost Status as a Quality Signal

The statistical analysis suggests that superhost listings have different review score distributions from non-superhost listings. This supports using superhost status as one signal of listing quality, while still considering other review and operational metrics.

Recommendation 6: Use Listing Segments for Targeted Market Actions

The K-Means segmentation separates listings into behavioural supply groups. Platform or market teams could use these segments to prioritise action. For example, low-rating low-activity listings may need quality improvement support, while established high-review listings can be analysed to understand strong performance patterns.

Recommendation 7: Use Review Themes to Support Host Improvement

The NLP analysis shows that guests often mention location, cleanliness, host quality, comfort, and ease of stay. Host-success teams could use these themes to guide listing improvement advice and operational checklists.

Recommendation 8: Avoid Unsupported Price Conclusions

Because the Melbourne dataset did not contain usable price values, price and revenue conclusions should not be made from this dataset. Future work should use a city or dataset snapshot with valid price data if pricing analysis is required.

13. Assumptions and Decision Log

The full assumptions and decision log are documented in:

```
reports/decision_log.md
```

Major decisions include:

- Selecting one city for depth.
- Using Python for pipeline development.
- Using DuckDB for local analytical modelling.
- Excluding raw data from Git.
- Including report outputs for reviewer evidence.
- Pivoting away from price analysis because the Melbourne price fields were missing.

- Using non-parametric statistical tests.
- Using K-Means clustering for explainable unsupervised listing segmentation.
- Using lightweight NLP instead of external LLM APIs to keep review analysis reproducible and auditable.

14. Limitations and Caveats

Important limitations include:

- The project analyses only Melbourne.
- The dataset is a public scraped snapshot and may contain missing or inconsistent values.
- Price fields were missing, so price and revenue analysis were not performed.
- Calendar unavailable days are only an occupancy proxy.
- Review frequency is only a proxy for demand.
- Statistical tests do not prove causation.
- K-Means segmentation is exploratory and should not be treated as a production classification model.
- The sentiment analysis uses a simple lexicon-based method, so it should be interpreted as a directional indicator rather than full human-level sentiment classification.
- Advanced production-grade machine learning, transformer-based NLP, LLM applications, and model deployment were not completed due to prioritisation.

15. Future Improvements

With more time, the project could be extended by:

- Adding another city with complete price data.
- Building a Streamlit dashboard.
- Adding Docker for reproducibility.
- Implementing dbt models and tests.
- Adding machine learning price prediction using a dataset with valid price fields.
- Extending the current NLP sentiment analysis with topic modelling, transformer-based sentiment models, or review-quality classification.
- Comparing K-Means segmentation with DBSCAN or hierarchical clustering.
- Deploying the pipeline to a cloud platform.
- Adding automated data quality tests.
- Building a geographic map using neighbourhoods.geojson.
- Adding a monitoring layer to track changes in future Inside Airbnb snapshots.

16. Reflection

This project was scoped to prioritise core data engineering and analytical quality.

The main trade-off was choosing depth over breadth. Instead of attempting every optional section superficially, the project focused on building a reproducible pipeline, clean outputs, SQL analysis, EDA, statistical testing, ML

segmentation, NLP review analysis, and clear reporting.

A major learning was that real-world public datasets may not support the original analytical plan. In this case, missing price data required a responsible pivot from price analysis to supply, availability, reviews, host concentration, quality signals, segmentation, and review themes.

The most important professional decision was to avoid inventing or imputing unsupported price values. This improves trust in the final analysis.

The ML and AI/NLP additions were intentionally kept explainable and reproducible. This was more appropriate for the assessment than using complex methods that would be harder to validate within the project timeline.

Appendix A. AI Usage Disclosure

The AI usage disclosure is documented in:

```
reports/ai_usage_disclosure.md
```

AI assistance was used for project planning, code structure guidance, debugging support, README drafting, and report structure guidance. All outputs were reviewed, executed, and validated locally.

Appendix B. Figure Gallery

Figure B.1: Availability By Room Type

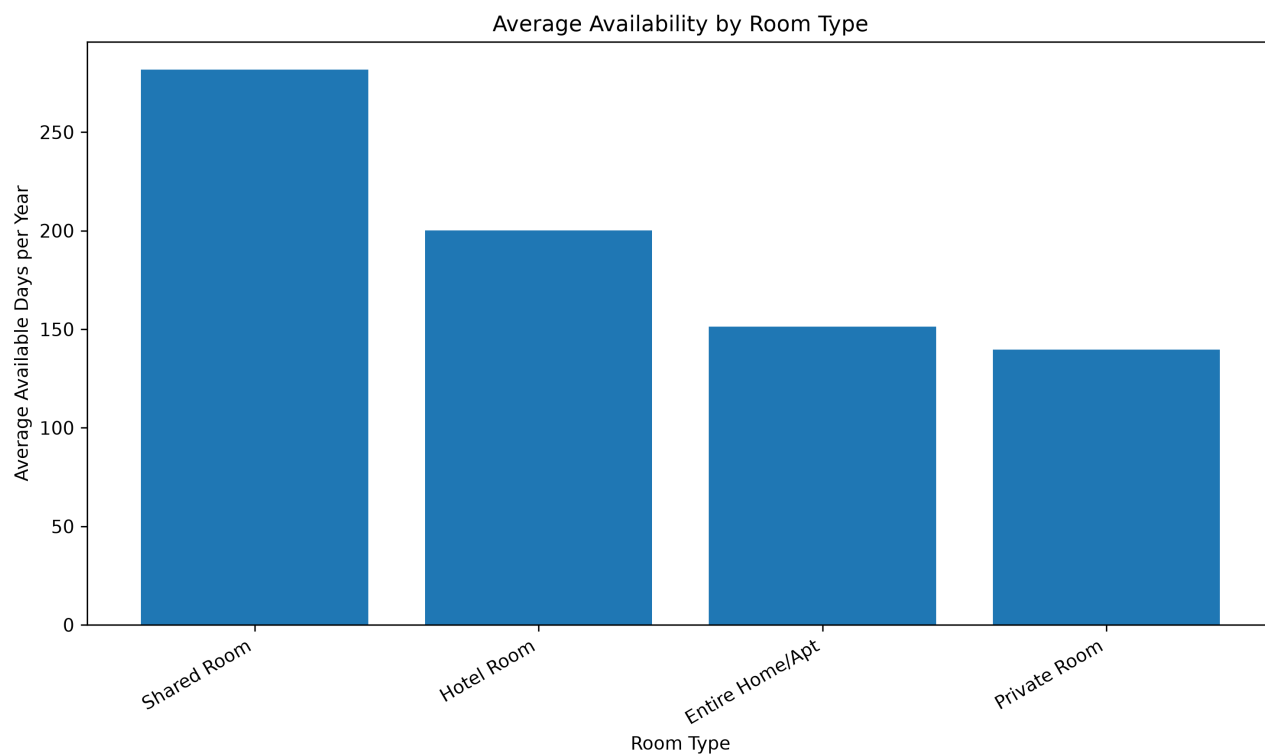


Figure B.2: Listing Segments Pca

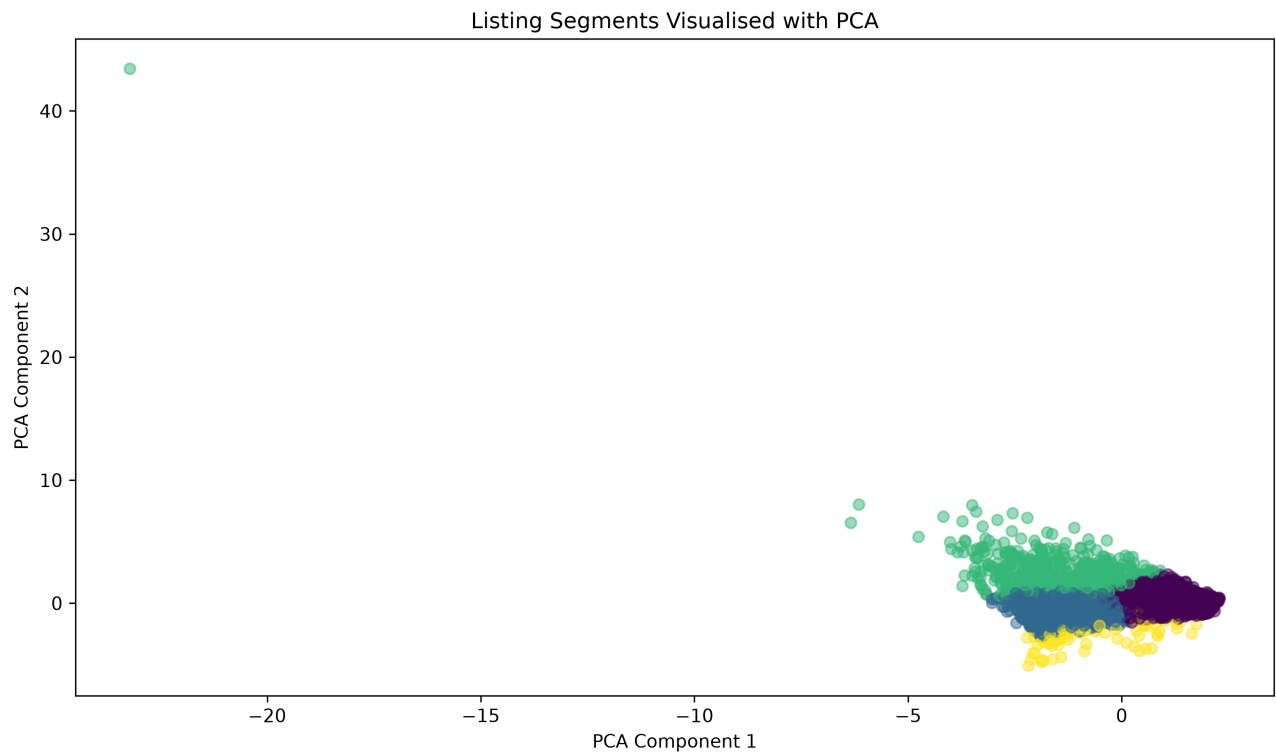


Figure B.3: Listing Supply By Room Type

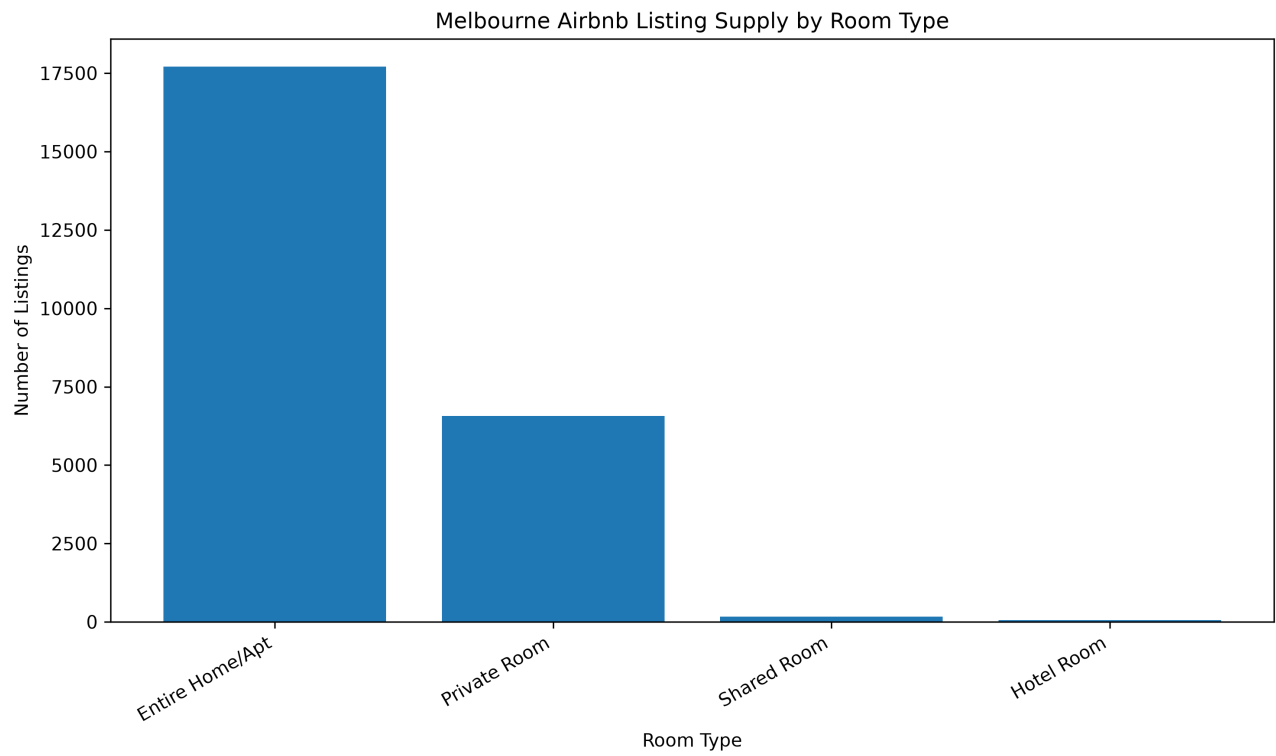


Figure B.4: Monthly Availability Rate

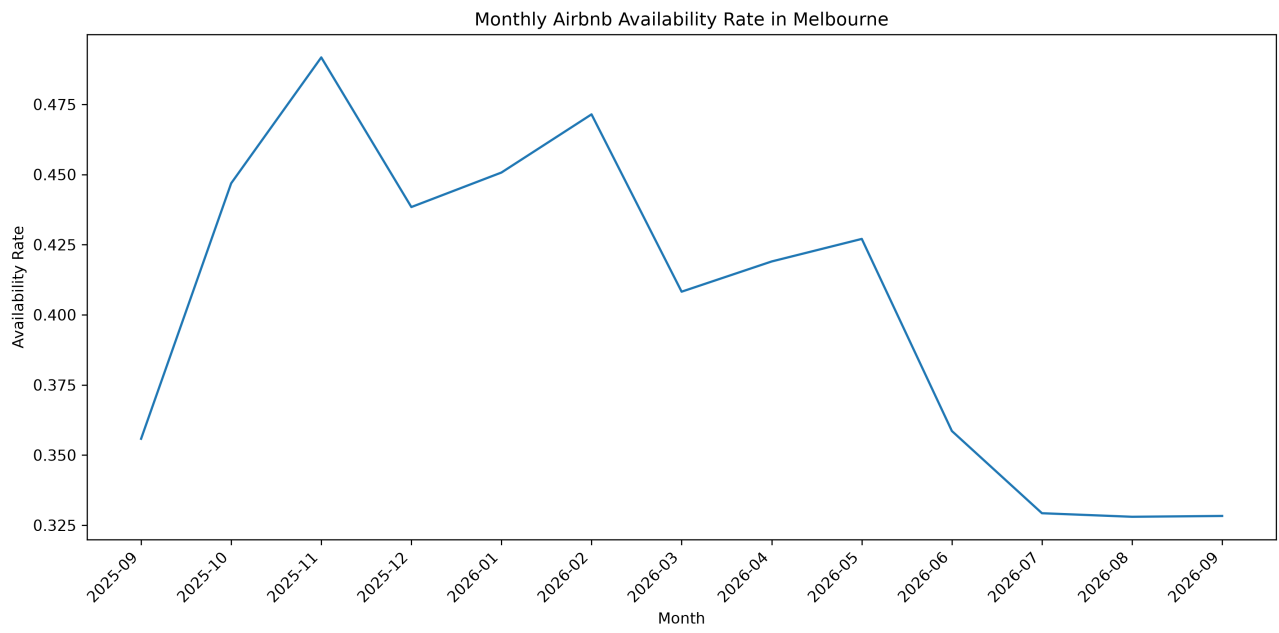


Figure B.5: Review Score Distribution

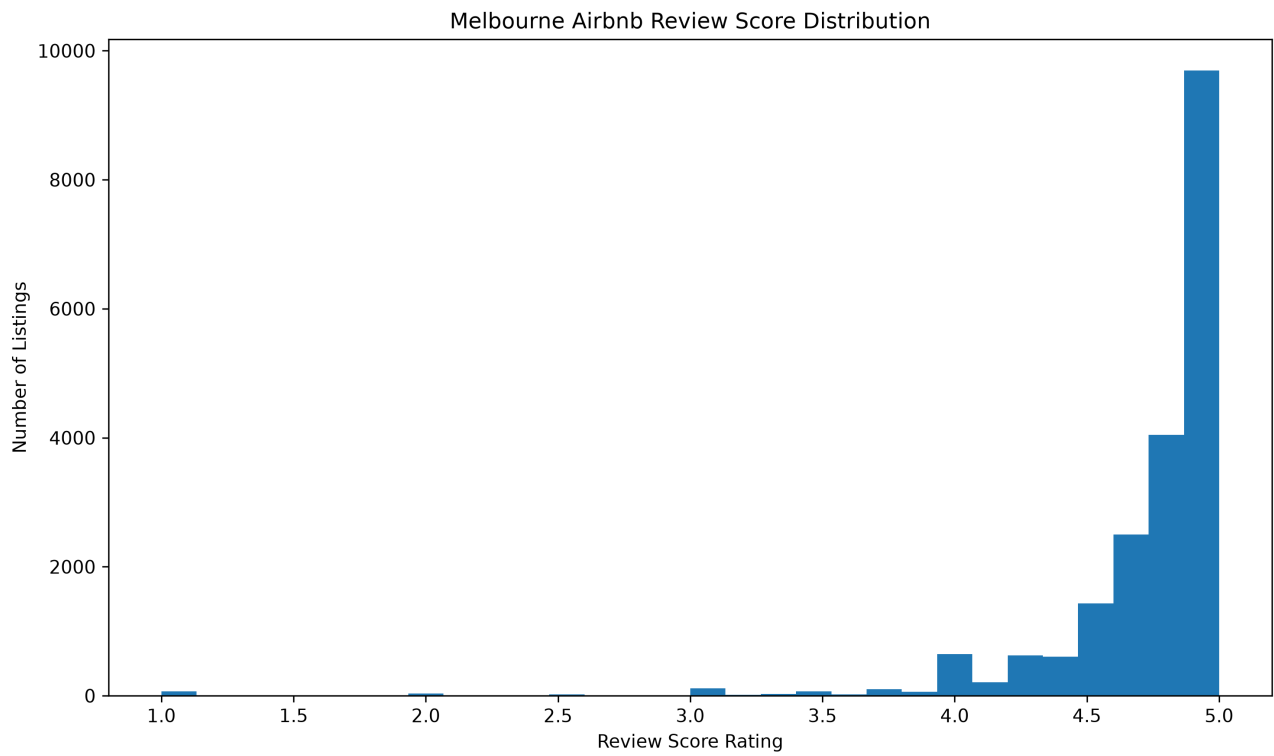


Figure B.6: Review Sentiment Distribution

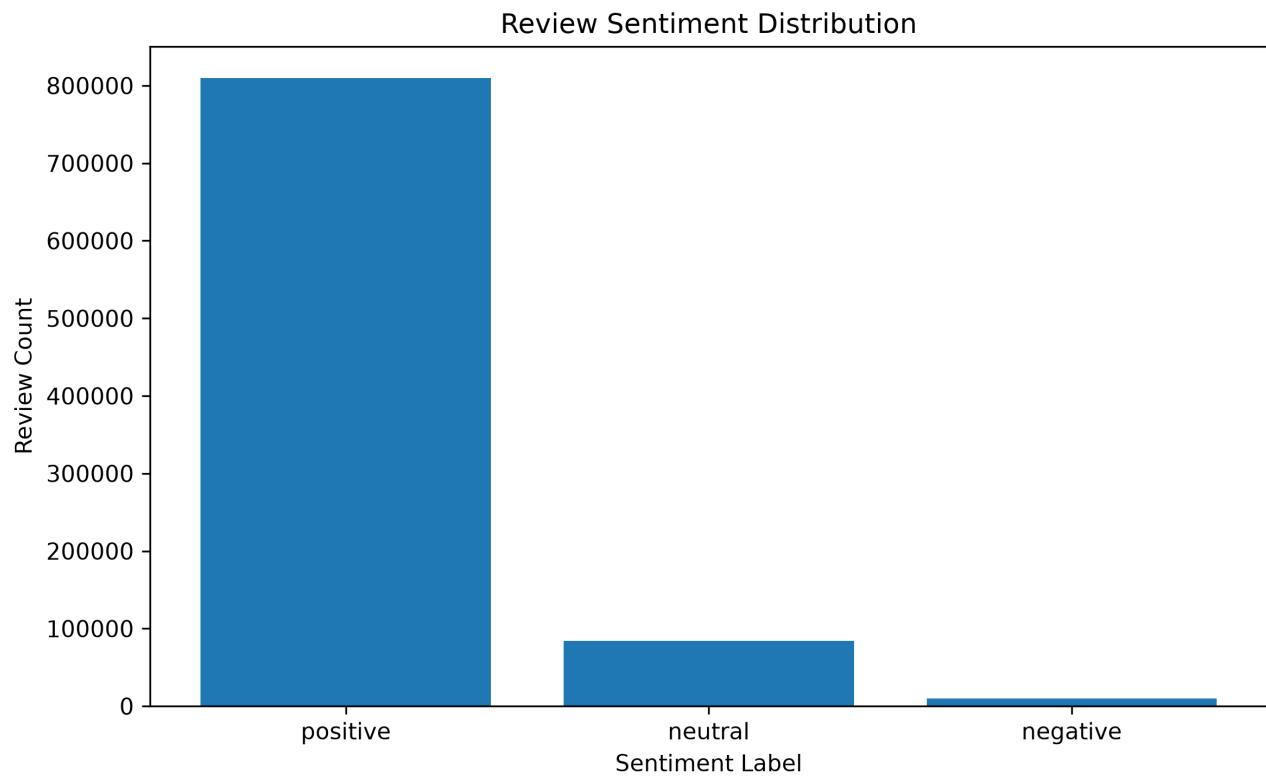


Figure B.7: Reviews By Room Type

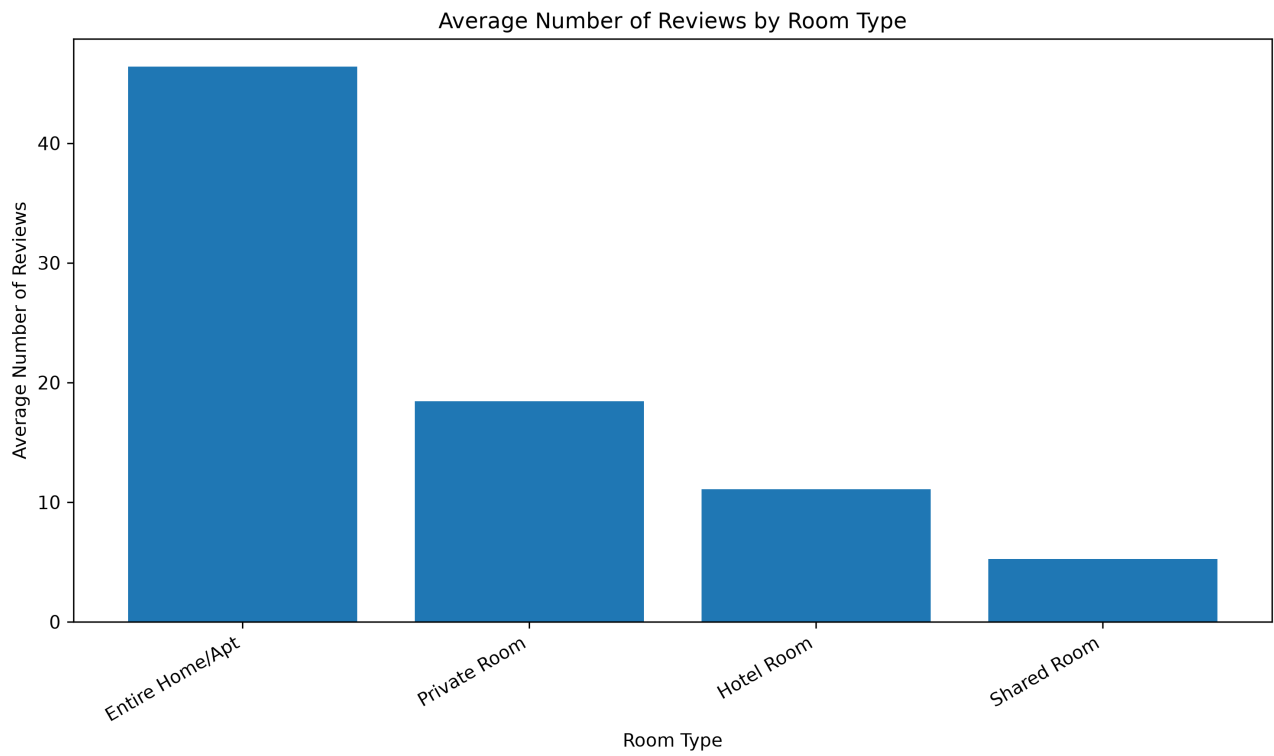


Figure B.8: Segment Profile Availability

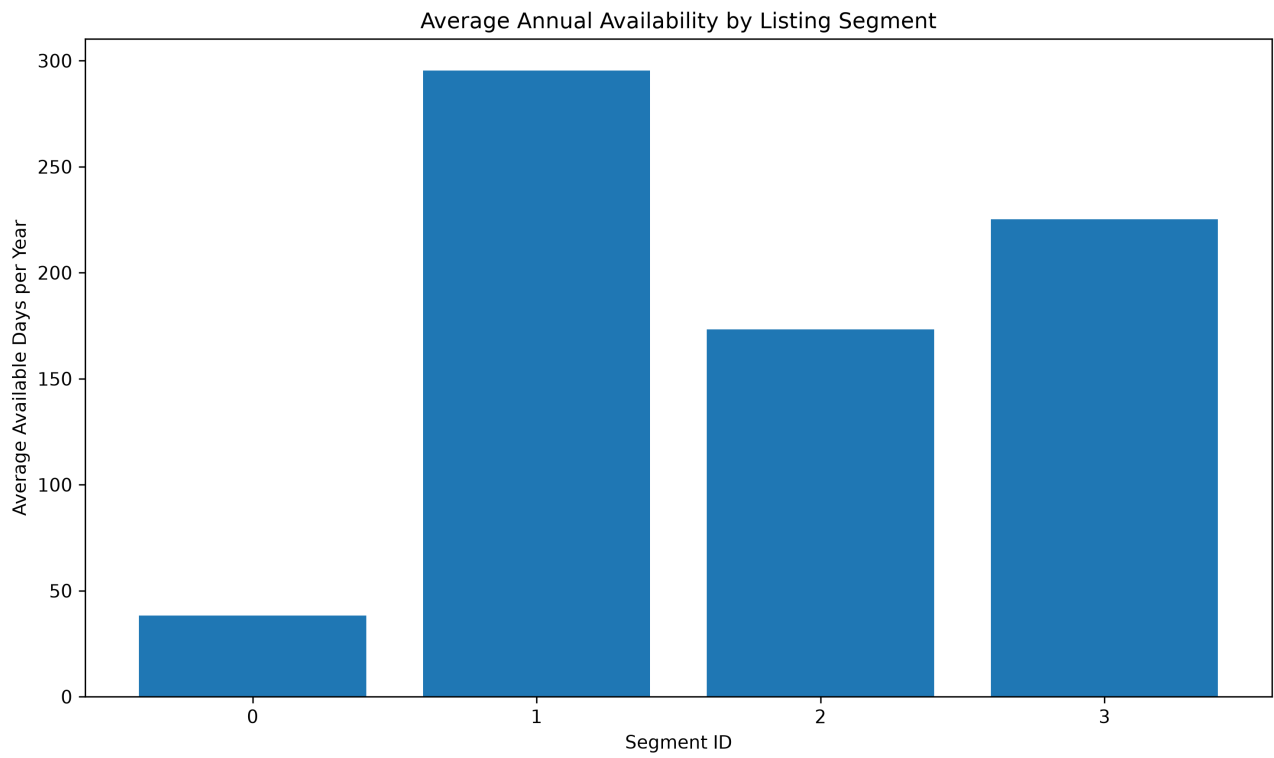


Figure B.9: Top Hosts By Listing Count

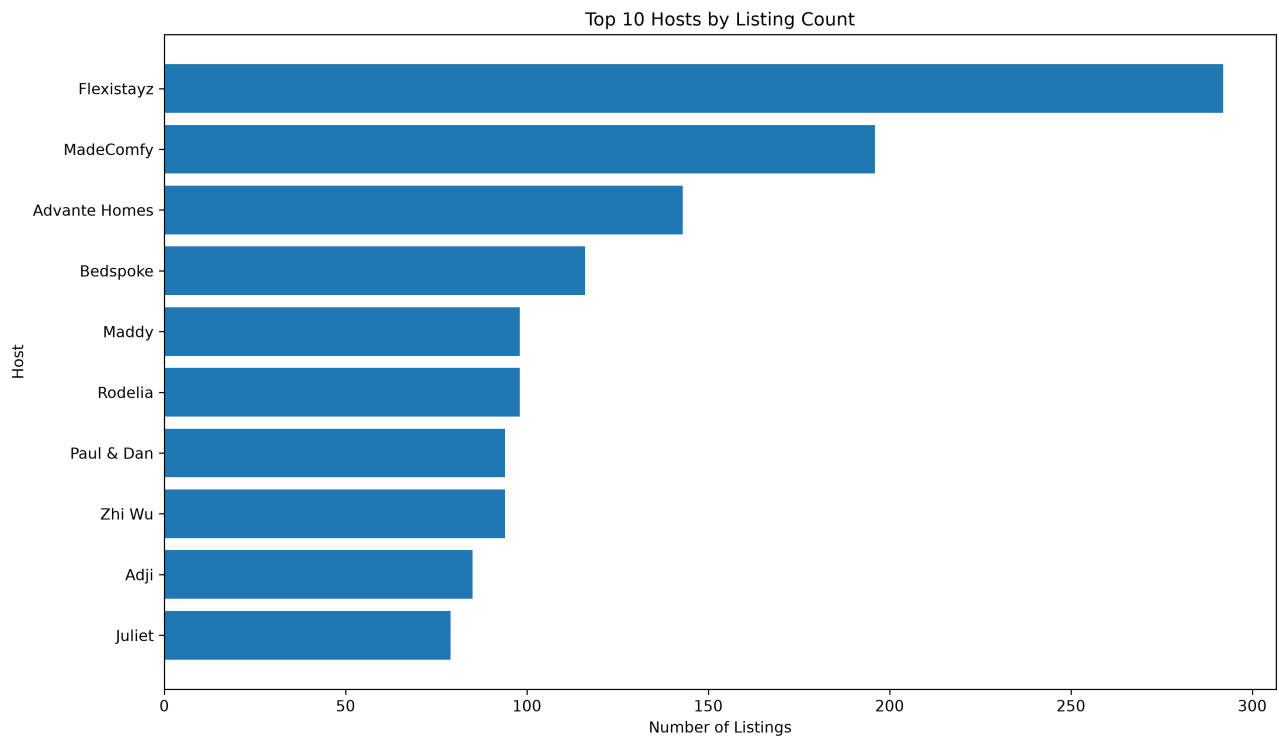


Figure B.10: Top Neighbourhoods By Listing Count

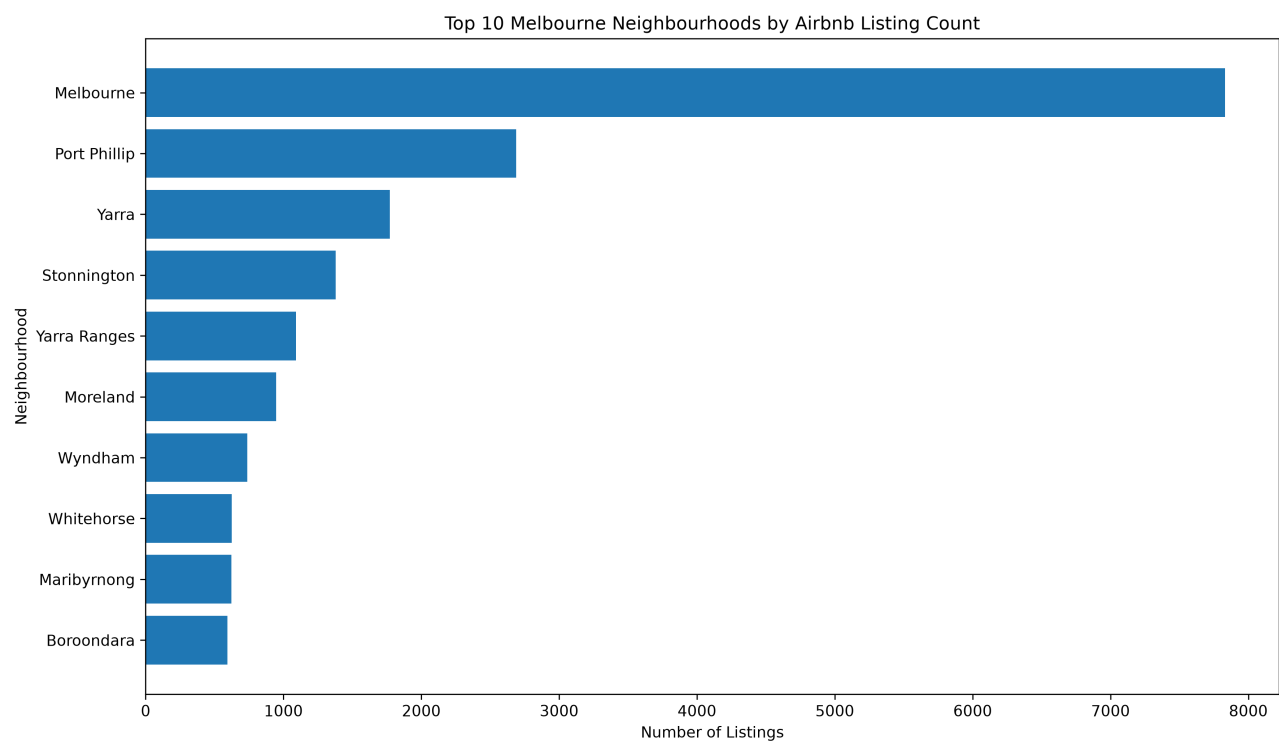
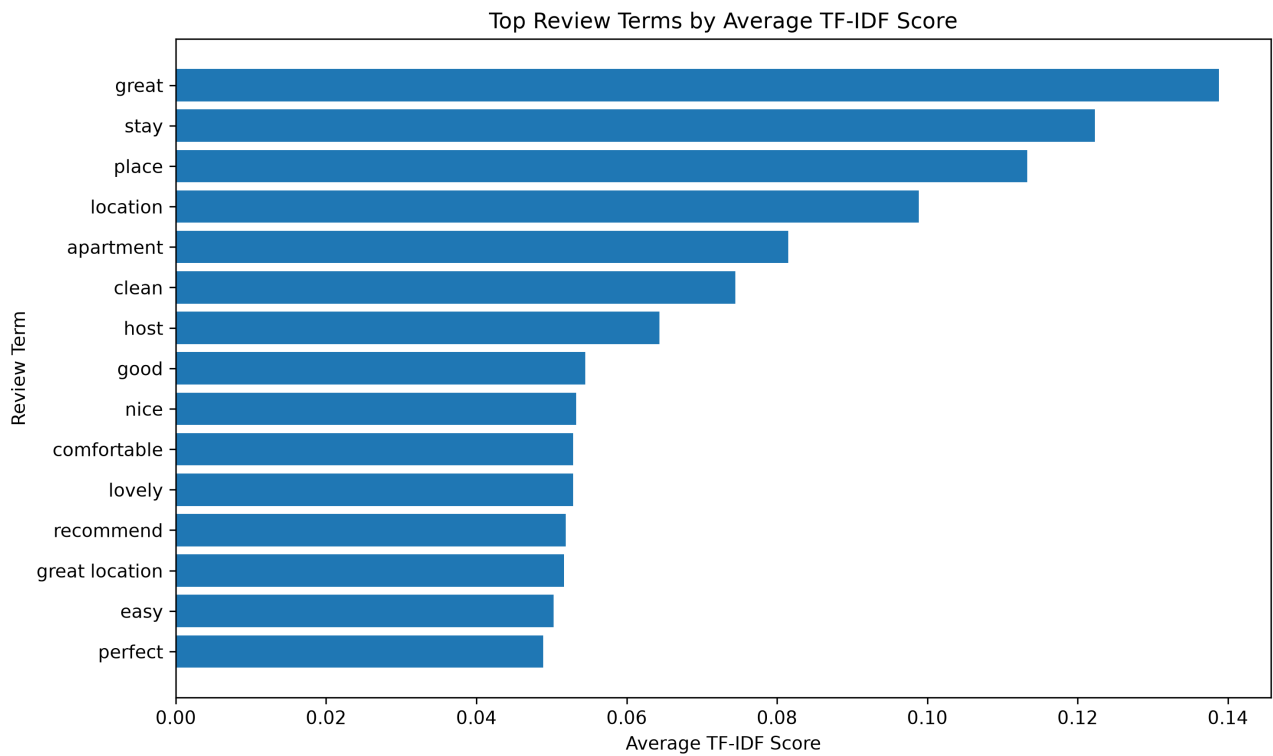


Figure B.11: Top Review Terms



Appendix C. SQL Output Tables

C.1: Query 01 Output

Source file: reports/sql_outputs/query_01_output.csv

room_type	listing_count	average_availability_365	median_availability_365	average_number_of_reviews
Entire Home/Apt	17703	151.37	130.0	46.39
Private Room	6563	139.74	89.0	18.45
Shared Room	170	281.64	364.0	5.25
Hotel Room	55	200.16	255.0	11.07

C.2: Query 02 Output

Source file: reports/sql_outputs/query_02_output.csv

neighbourhood_cleansed	listing_count	average_availability_365	average_number_of_reviews	average_review_score
Melbourne	7832	140.12	49.14	4.67
Port Phillip	2687	127.79	33.57	4.79
Yarra	1771	112.48	44.83	4.81
Stonnington	1380	123.04	31.37	4.78
Yarra Ranges	1092	226.94	82.29	4.85
Moreland	946	114.89	29.79	4.76
Wyndham	737	203.67	18.0	4.72
Whitehorse	624	158.19	18.76	4.67

C.3: Query 03 Output

Source file: reports/sql_outputs/query_03_output.csv

neighbourhood_cleansed	listing_count	average_occupancy_rate_estimated	average_availability_365	average_number_of_reviews
Darebin	584	0.7066	106.91	27.6
Yarra	1771	0.6918	112.48	44.83
Moreland	946	0.6836	114.89	29.79
Stonnington	1380	0.6623	123.04	31.37
Port Phillip	2687	0.6499	127.79	33.57
Boroondara	593	0.6314	134.53	24.26
Melbourne	7832	0.616	140.12	49.14
Glen Eira	527	0.6018	145.33	22.52

C.4: Query 04 Output

Source file: reports/sql_outputs/query_04_output.csv

host_is_superhost	listing_count	average_review_score	median_review_score	average_number_of_reviews
True	6781	4.87	4.9	78.59
	793	4.72	4.8	38.6
False	12735	4.66	4.79	29.86

C.5: Query 05 Output

Source file: reports/sql_outputs/query_05_output.csv

is_weekend	calendar_records	availability_rate
True	2547064	0.4019
False	6392151	0.412

C.6: Query 06 Output

Source file: reports/sql_outputs/query_06_output.csv

host_id	host_name	listing_count	average_availability_365	average_number_of_reviews
90729398	Flexistayz	292	94.66	5.8
279001183	MadeComfy	196	238.79	17.18
343442154	Advante Homes	143	93.47	22.69
446080599	Bedspoke	116	324.56	38.03
392306676	Rodelia	98	333.86	4.08
38814953	Maddy	98	328.31	5.2
244502930	Zhi Wu	94	278.09	11.67
1739996	Paul & Dan	94	237.94	191.37

C.7: Query 07 Output

Source file: reports/sql_outputs/query_07_output.csv

room_type	listing_count	average_number_of_reviews	average_reviews_per_month	average_availability_365	average_review_score
Entire Home/Apt	17703	46.39	1.48	151.37	4.74
Private Room	6563	18.45	0.73	139.74	4.72
Shared Room	170	5.25	0.53	281.64	4.46
Hotel Room	55	11.07	0.17	200.16	4.51

Appendix D. EDA Output Tables

D.1: Availability By Room Type

Source file: reports/eda_outputs/availability_by_room_type.csv

room_type	listing_count	average_availability_365	median_availability_365
Shared Room	170	281.6353	364.0
Hotel Room	55	200.1636	255.0
Entire Home/Apt	17703	151.3659	130.0
Private Room	6563	139.7355	89.0

D.2: Listing Supply By Room Type

Source file: reports/eda_outputs/listing_supply_by_room_type.csv

room_type	listing_count	average_availability_365	median_availability_365	average_number_of_reviews
Entire Home/Apt	17703	151.3659	130.0	46.3862
Private Room	6563	139.7355	89.0	18.4469
Shared Room	170	281.6353	364.0	5.2529
Hotel Room	55	200.1636	255.0	11.0727

D.3: Monthly Availability Rate

Source file: reports/eda_outputs/monthly_availability_rate.csv

month	calendar_records	availability_rate
2025-09	465329	0.3558
2025-10	759221	0.4469
2025-11	734730	0.4917
2025-12	759221	0.4384
2026-01	759221	0.4507
2026-02	685748	0.4714
2026-03	759221	0.4082
2026-04	734730	0.419

D.4: Reviews By Room Type

Source file: reports/eda_outputs/reviews_by_room_type.csv

room_type	listing_count	average_number_of_reviews	median_number_of_reviews	average_reviews_per_month
Entire Home/Apt	17703	46.3862	15.0	1.4842
Private Room	6563	18.4469	3.0	0.7257
Hotel Room	55	11.0727	5.0	0.1658
Shared Room	170	5.2529	1.0	0.5283

D.5: Top Hosts By Listing Count

Source file: reports/eda_outputs/top_hosts_by_listing_count.csv

host_id	host_name	listing_count	average_availability_365	average_number_of_reviews
90729398	Flexistayz	292	94.6644	5.8048
279001183	MadeComfy	196	238.7857	17.1837
343442154	Advante Homes	143	93.4685	22.6853
446080599	Bedspoke	116	324.5603	38.0345
38814953	Maddy	98	328.3061	5.2041
392306676	Rodelia	98	333.8571	4.0816
1739996	Paul & Dan	94	237.9362	191.3723
244502930	Zhi Wu	94	278.0851	11.6702

D.6: Top Neighbourhoods By Listing Count

Source file: reports/eda_outputs/top_neighbourhoods_by_listing_count.csv

neighbourhood_cleansed	listing_count	average_availability_365	average_number_of_reviews	average_review_score
Melbourne	7832	140.1196	49.1398	4.6652
Port Phillip	2687	127.7949	33.569	4.7905
Yarra	1771	112.4828	44.8255	4.8113
Stonnington	1380	123.0406	31.3703	4.7802
Yarra Ranges	1092	226.9386	82.2921	4.8486
Moreland	946	114.8932	29.7907	4.7553
Wyndham	737	203.673	18.0014	4.7156
Whitehorse	624	158.1907	18.7596	4.6654

Appendix E. Statistical Output Tables

E.1: Statistical Tests Summary

Source file: reports/statistical_outputs/statistical_tests_summary.csv

hypothesis	test_used	p_value	effect_size	business_interpretation
H1: Entire-home listings have different annual availability than private rooms	Mann-Whitney U test	0.0	0.0659	This tests whether different accommodation types behave differently in terms of yearly availability. A significant result suggests supply strategy differs by room type.
H2: Superhost listings have different review scores than non-superhost listings	Mann-Whitney U test	0.0	0.259	This tests whether superhost status is associated with guest satisfaction. A significant result suggests superhost status may be a useful quality signal.
H3: Annual availability differs across neighbourhoods	Kruskal-Wallis test	0.0	0.0565	This tests whether supply availability patterns differ by location. A significant result suggests neighbourhood-level market monitoring is needed.
H4: Weekend availability differs from weekday availability	Mann-Whitney U test	0.0	-0.0101	This tests whether availability patterns differ between weekends and weekdays. A significant result may indicate seasonal or demand-driven booking behaviour.